

AKSHAT SINHA

Front-end and DevOps Engineer

📍 Vellore, Tamil Nadu | 📞 +91 9142812513 | ✉ akshat.sinha2022@vitstudent.ac.in | 📄 LeetCode | 📁 GitHub | 🔗 LinkedIn | 🏆 Codeforces

EDUCATION

Vellore Institute of Technology, Vellore Campus, Tamil Nadu
B.Tech in Computer Science Core

🏠 Pursuing (Expected May 2026)
Current CGPA **8.88/10**

Delhi Public School Ranchi

Class XII (Senior Secondary) Percentage: **89.6%** | Class X (Secondary) Percentage: **95%**

TECHNICAL & DESIGN SKILLS

- **Programming & Data:** C++, Java, Python, PWSH, ASP.NET MVC, jQueryUI, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib
- **Web & Tools:** React, Node.js, MySQL, Mongo, Git, Jupyter, Docker, Kubernetes, Jenkins, Ansible, Terraform, CircleCI, Excel, Figma, Blender
- **Cloud Native Ecosystem:** AWS, VMware IaC Architecture, Intel SGX enclaves (concepts), Cloud OTA Firmware, Cisco ACI, Chaos, CI/CD
- **Soft Skills:** Team Collab (cross-functional), Leadership, Documentation, Risk & Project Management, Agile (Scrum), Business modelling

RELEVANT COURSEWORK

OS, DBMS, DSA, DAA, COA, OOP, CN, Compiler Design, Data Mining & Analytics, SWE Principles, ML & AI, IOT, Embedded Systems

EXPERIENCE

INSYD (January 2025 – June 2026)

Remote | Software Engineer Intern

Technologies: Next.js, Convex, AWS, PostHog, WebSockets, GitHub/GitLab

- Developed and shipped features for **5 internal products** — **Kard**, **Ello**, **DesignLens**, **StudioOS**, and **Dais** — on a **Next.js**, **Convex**, and **AWS** stack with **Mastra** AI agents, **PostHog** analytics, **CI/CD** pipelines, and live **WebSocket** integrations
- Revamped 3 websites as **proof of concept** demos targeting **UI and UX** redesign and **technical and operational** modernisation for prospective client acquisition
- Developed **sales and distribution ecosystem** for **Novara Locking Solutions** (hardware partner), building core features and user interface currently being tested by **10 to 15** field salesmen for feedback

PISTAR TECH (September 2025 – December 2025)

Hybrid | Python Developer Intern

Technologies: Python (FastAPI, Flask), Gunicorn, Uvicorn, BeeWare, Buildozer, Chaquopy, MATLAB GUI, HTML, SCSS, TypeScript, JavaScript

- Develop **MATLAB GUI** and **Python ASGI services** for **PodiaScanner Mat** (plantar-pressure) and **Smart Chair** (AI posture monitoring), resolving **visualization issues** (foot orientation, heatmap accuracy) improving **diagnostic accuracy by 35%**, implementing **CSV export** (1024 sensors), **patient management** (BMI, report IDs), and **biomechanical analysis** (arch index, pressure distribution) reducing **report generation by 50%**; deploy via **FastAPI/Flask** on **Uvicorn**, package with **BeeWare**
- Optimize websites (90%+ Lighthouse) and web apps for **Core Web Vitals (LCP, CLS, INP)**, improving load time by 38%, interaction latency by 27%, and layout stability by 22%, enhancing **UI/UX**, **performance**, **testable code**, and **incremental deployments** for fast and stable experiences

POINT-OF-CARE MICROFLUIDICS PRIVATE LIMITED (July 2025 – December 2025)

VIT Vellore | Onsite SWE Intern

Technologies: Full-stack Infra Delivery Ecosystem, paved-road CI/CD, GitOps, IaC, ASP.NET Core, Razor/Blazor, Wisej.net(Avalonia, Uno) 4.0, .NET MAUI, Node, Drogon, Nginx, Envoy, Containers, Kafka, JEE, Microprofile, Micronaut, Spring Boot, Android Fluoride, AWS, ClickHouse, OpenAPI, Confluent Cloud, IntelliJ, ELK

- Developed **production-ready e-commerce** and **lab-on-chip healthcare** (e.g., microarray patch) **platforms** with **Node + Drogon backends** and **ASP.NET Core frontends**, implementing **cross-platform UX**, live **device telemetry dashboards**, **Kafka event streaming**, **blue/green** and **canary deployments**, and **automated rollback**, reducing **error rates by 10%** and maintaining **P95 latency under 250ms**
- Engineered **PCI-aware checkout flows**, **tokenized payments**, analytics and **microservices architecture** (**Micronaut-Spring Boot & JEE**) exposing **REST/GraphQL APIs**, **webhooks**, and **SPARQL endpoints**, blocking **50% of fraudulent attempts**, cutting **payment failures by 50%** per merchant fingerprint, and enabling **zero-downtime releases** with traffic shaping and platform standardization

SCHOLARS MERIT ONLINE PRIVATE LIMITED (May 2025 - July 2025)

Noida, UP | Full Stack MERN Development Intern

Technologies: React.js, Node.js, Express.js, MongoDB, Vaadin, LAMP Stack, test suite, YARP, Katalon Studio

- Implemented and maintained **MERN apps** with **responsive UIs** using **React.js & Vaadin**, integrating **RESTful APIs** via **Node.js/Express.js**
- Improved **database performance** (**MongoDB, LAMP**), resolved **critical bottlenecks**, and contributed to **testing, code reviews**

PROJECTS

Real-Time Chess Cheat Detection Engine

<https://github.com/akshatsinhao/chess-cheat-detect.git>

Technologies: Python, Stockfish, OpenCV, TensorFlow, Scikit-learn, Selenium, Docker, MySQL, Isolation Forest

- Constructed a **real-time chess cheat detection engine** using **Python**, **Stockfish NNUE**, **OpenCV**, **TensorFlow**, **Scikit-learn**, **Selenium**, **Docker**, and **MySQL**, achieving **98% move analysis accuracy** and **30% fewer false positives** through **Isolation Forest based anomaly detection**, **feature engineering** including session entropy, think-time, and opening adherence, and **scalable containerized deployment**

TakesTakesTakes — Two-Player Chess Game Website

<https://takestakestakes.netlify.app/>

Technologies: HTML, CSS, JavaScript, Node.js, WebSockets, MongoDB, Supabase, Cypress, OpenGL, Instrumented Code

- Built a **real-time 2P chess site** with **Vite**, **Chessboard.js**, **WebSockets**, **MongoDB**, **Supabase**, including **auth**, **game sync**, **profiles**, **history**

FlowDiag: Program or Code Function Visualization Extension

<https://github.com/akshatsinhao/flowdiag.git>

Technologies: JavaScript, TypeScript, Yeoman

- Architected a **VS Code extension** in **JavaScript/TypeScript** using **Tree-sitter** and **Mermaid.js** to parse **function ASTs** and generate **interactive flowcharts**; streamlined **code comprehension** with async **WASM-backed, cross-platform Node.js support**

Bharat KYC (LokSakshya)

<https://github.com/akshatsinhao/KYC-App.git>

Technologies: Android/Kotlin, WorkManager, Android Keystore (AES-GCM), Zip4j, BouncyCastle, Dynamic Feature Modules, Node, OpenAPI 3.0, OkHttp, JUnit

- Built an **offline-first Android KYC app/SDK** (Kotlin, minSdk 23) with **Dynamic Feature Modules** (base APK **<=15 MB**); implemented **on-device Aadhaar Paperless Offline eKYC** using **Zip4j** and **BC (PKCS#7/X.509)**, **lightweight OCR parsers** (**PAN, DL, Voter ID**), **face liveness/match**; enforced **privacy-by-design** with **AES-GCM** (Android Keystore), **30-day retention**, **zero PII egress**

- Delivered **OpenAPI 3.0 Node.js/Express stub** and re-implemented **on-prem KYC stub in Go (Fiber)** reducing **p99 latency 40%** with **<40 MB RAM**; shipped **minimal SDK** (createSession, enqueueFinalize with **resultDigest/idempotency**) and **encrypted WorkManager sync queue (OkHttp, exponential backoff)**; optional **on-device assistant (TFLite)**, local analytics, JUnit tests

AWARDS & ACHIEVEMENTS

Winner, Convex Components Authoring Challenge (2025)

Convex Components Directory

Won **prize money** for **two open-source components** approved into the **official Convex directory** and **published to npm**:

convex-cascading-delete (declarative cascading deletes with atomic inline & batched scheduled modes, zero schema migration) and

convex-api-keys (native, zero-dependency API key management with **SHA-256 hashing**, **sliding-window rate limiting**, **RBAC**, and real-time analytics — first entry in the API Usage category)

Winner, Ideathon 2.0 by Pixelate, VIT Vellore (March 2025)

Led **Team Rudiment** to **1st place** in Pixelate Ideathon 2.0 with **Authentica**, a **blockchain platform** for securing **digital content**, showcasing **innovation** and **teamwork** in tackling **content security** and **AI replication challenges**